How Many Runs Does an RL Post-Training Comparison Need? Forecasting Run-to-Run Spread from a Single Run

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Abstract

Reinforcement learning post-training is compared from single runs, and how many runs a comparison needs is set by a quantity nobody measures: how much a rerun would move the score. The usual recourse is to fix the run count in advance, or to spend runs estimating the spread before the comparison can begin, because after one run per arm there is no empirical spread to put in a stopping rule.

We forecast it instead. Training noise is not accumulated: a perturbation entering at one update is reweighted by every update that follows, so the covariance of a run about its mean obeys $S_{t+1} = A_t S_t A_t^\top + Q_t$. Carrying the gradient of the reported metric backwards along one stored trajectory turns that recursion into the across-seed variance of the metric, as a sum of scalars with no covariance matrix formed. On enumerable policies, where every input is exact, and frozen before the remaining seeds existed, it has a median error of 1.10× over 192 settings whose runs gain at most 0.036 in pass rate. Where runs learn it fails on that tier too, and seeds simulated in a model linearised at the start, with no run trained, are within 1.5× in 74.9% of settings.

Used to size comparisons whose ordering is clear, with every stopping threshold chosen on other tasks before it is applied, it reaches a 0.1% mistake rate in 3.14 runs, 5.6 to 8.2× fewer than a fixed budget measured below its cap and 2.12× fewer than estimating the deviation from spent seeds. A closed form that needs no run comes within 1.11× once its threshold is learned on other tasks; the stored run's scale replaces that learning. Near-ties are settled by no policy within the seeds we have.

On pretrained models accuracy is regime-dependent. Across 4 pre-specified regimes on Qwen2.5-0.5B the frozen forecast misses by factors of $1.66, 2.13, 2.80, 1.24$, inside the 4.2× relative error at which, with z set in sample, the procedure stops paying. On a task where most runs learn to answer one constant it is short by a factor of 25, and at 7B under Adam by 3.1 to 17×. We report both.

1 Introduction

Reinforcement learning with verifiable rewards is the standard way to turn a pretrained model into a reasoning one [DeepSeek-AI et al., 2025], and a paper reports that a recipe moved a benchmark by two points. Whether that is a result depends on a number nobody has: how much the same recipe would have moved it under a different seed. The problem is documented [Henderson et al., 2017, Hochlehnert et al., 2025], and the standard instrument for it is replication, which at RL post-training cost is close to unaffordable.

The practical form of the problem is not reporting but budgeting. Deciding that one recipe beats another means training seeds of each until the lead is convincing, and how many that takes is governed by the across-seed deviation. A practitioner who does not have it either fixes the seed count in advance, or spends seeds estimating it before the comparison can start. Both are expensive, and the second is expensive for a structural reason: after a single run per arm there is no empirical deviation at all, so the first two runs of every arm buy no evidence about the question being asked.

This paper forecasts that deviation from one run, and then spends it. The route runs through a correction to the obvious model of where run-to-run spread comes from. That model says gradient noise perturbs each update, perturbations accumulate, two runs execute a random walk away from each other, and reproducibility decays with run length. It is wrong. Noise from one update does not survive to the end of training unchanged: it is passed through every update that follows, which reweights it. What reaches the end is a convolution of injected noise with the sensitivity of the remaining trajectory, an object that obeys a recursion and that can be carried against the gradient of the reported metric rather than being formed.

Contributions.

- **An error bar from one run** (Theorem 2). Car-

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rying the metric gradient backwards along a single stored trajectory gives the across-seed variance of the reported score as a sum of scalars, one per update, with no covariance matrix formed. The transpose a backward pass would need is free, because the mean update of any count-based estimator is the gradient of a scalar potential and its Jacobian is therefore symmetric. On enumerable policies, where every input to the forecast is exact, and frozen before the remaining seeds were trained, its median error is 1.10× across 192 settings of at most 25 updates.

- **Comparisons sized from one run** (Section 6). With every stopping threshold chosen on other tasks before it is applied, the forecast settles 1020 comparisons at a 0.1% mistake rate in 3.14 runs, against 14 for an uncapped fixed budget and 2.12× as many for a deviation estimated from spent seeds. A closed form that needs no run does nearly as well once its threshold is learned (1.11× [0.97, 1.28]); what the stored run adds is an absolute scale, which lets the textbook threshold be used with no calibration at all.
- **The condition under which it pays.** Corrupting the forecast deliberately puts the break-even against a fixed budget at 4.2× relative error; a pure scale error is free only where the threshold can be tuned on comparisons with known outcomes.
- **An honest boundary** (Section 7). On Qwen2.5-0.5B accuracy is regime-dependent: across 4 pre-specified regimes the errors are $1.66, 2.13, 2.80, 1.24$, all inside the break-even above. Outside it, on a task where most runs learn to answer one constant whatever they are asked, the forecast is short by 25×, and at 7B under Adam the forecast is short by between 3.1 and 17. The shortfall reappears on enumerable policies once they learn; there, seeds simulated in the tangent model (Section 3.2) recover the spread in 74.9% of settings with no run trained.

Every experiment runs on one laptop, and every pretrained forecast spans at most 12 updates with LoRA adapters, so none of it establishes behaviour at frontier scale (Section 9).

2 Setup

A policy $\pi_\theta(y \mid x)$ answers a prompt $x \sim \mathcal{D}$ and a verifier returns a binary reward $r(x, y) \in \{0, 1\}$. The objective is $J(\theta) = \mathbb{E}_x[p(x)]$ with $p(x) := \Pr[r = 1 \mid x]$. Write $\boldsymbol{s}(y) := \nabla_\theta \log \pi_\theta(y \mid x)$, so $\mathbb{E}_y[\boldsymbol{s}] = 0$, and let $u_r := \mathbb{E}[\boldsymbol{s} \mid r]$. Because the reward is binary, every quantity below depends on the policy only through the per-prompt objects (p, u_1, S_0, S_1), where $S_r := \mathrm{Cov}(\boldsymbol{s} \mid r)$, which is what makes an exact finite-G treatment possible.

A batch is P prompts drawn from $\mathcal{D}$ with G responses each; $R = PG$ is fixed by hardware. With k the number of successes in a group we consider any advantage of the form

$$A_j = w(k, r_j), \tag{1}$$

that is, any rule in which a response's advantage depends only on its own reward and on how many of its group-mates succeeded. It covers RLOO [Ahmadian et al., 2024], GRPO with a mean baseline [Liu et al., 2025], standardised GRPO [Shao et al., 2024] and a fixed external baseline. The per-prompt and batch estimators are $z := G^{-1} \sum_j w(k, r_j) \boldsymbol{s}(y_j)$ and $\hat{\boldsymbol{g}} := P^{-1} \sum_i z_i$.

Two levels of noise. Prompts are sampled and then responses are sampled within each prompt, so by the law of total variance

$$\mathrm{Cov}(\hat{\boldsymbol{g}}) \;=\; \frac{1}{P}\Big[\,\Sigma_{\mathrm{b}} + \tfrac{1}{G-1}\Sigma_{\mathrm{w}}\,\Big], \tag{2}$$

with Σ_{b} the covariance across prompts of the per-prompt mean update and Σ_{w} the within-prompt term, written in the $(G-1)$ convention the leave-one-out estimators use. Both traces are recoverable from the microbatch gradients a trainer already accumulates. We write τ_{b} and τ_{w} for those traces.

Intervals. Seed experiments produce pairwise quantities: n runs give $\binom{n}{2}$ distances, which are not $\binom{n}{2}$ observations, since every run appears in $n-1$ of them. Treating them as independent understates the standard error by roughly a factor of two at the seed counts anyone trains. Every interval here resamples *runs* and rebuilds the pairwise quantity from the resampled set.

3 Noise is filtered, not accumulated

Fix a policy θ_0 and start two runs from it. They share parameters, optimiser state, data distribution and recipe, and independently draw everything the seed controls: which prompts each batch contains and which responses are sampled. That is the comparison a practitioner makes when they rerun, and it is the estimand throughout, so both terms of (2) contribute.

Write $\delta_t := \theta_t - \bar{\theta}_t$ for one run's departure from the mean trajectory. With updates $\theta \leftarrow \theta + \eta\hat{\boldsymbol{g}}$ and $\mathcal{J}_t := \partial\bar{\boldsymbol{g}}/\partial\theta$,

$$\delta_{t+1} = A_t\delta_t + \eta\,\xi_t, \qquad A_t := I + \eta\mathcal{J}_t, \tag{3}$$

to first order in δ, where $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$ and $\mathrm{Cov}(\xi_t) = \Sigma_t/P$.

If the objective were flat, $\mathcal{J} = 0$, this would be a random walk: divergence would accumulate as $\mathbb{E}\|\Delta_t\|^2 \propto t$ and a longer run would be strictly less reproducible. That is the implicit model behind measuring run-to-run variance by brute force. But two RLVR runs climb the same objective, and A_t reweights a perturbation rather than passing it through.

Theorem 1 (Finite-time covariance)**.** *Let $S_t := \mathrm{Cov}(\delta_t)$, with $S_0 = 0$ when runs start together. Under* (3)*,*

$$S_{t+1} = A_t S_t A_t^\top + Q_t, \qquad Q_t := \eta^2 \Sigma_t / P, \tag{4}$$

hence $S_T = \sum_{t<T} \Phi_{T,t+1} Q_t \Phi_{T,t+1}^\top$ with $\Phi_{T,s} := \prod_{u=s}^{T-1} A_u$ taken in order of decreasing u. Two independent seeds satisfy $\mathrm{Cov}(\theta_T^A - \theta_T^B) = 2S_T$.

Proof sketch. Taking the covariance of (3) leaves a cross term $\mathbb{E}[\delta_t \xi_t^\top]$. It vanishes without assuming ξ_t and δ_t are independent, which they are not, since the batch noise is drawn at θ_t and that parameter was produced by earlier noise. What holds is that ξ_t is a martingale difference, $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$, and δ_t is $\mathcal{F}_t$-measurable, so the cross term is zero by the tower rule. Unrolling from $S_0 = 0$ gives the sum. □

The factor of two belongs to the difference of two runs and not to the covariance of one; keeping them apart is what makes the statement unambiguous.

3.1 Carrying it to the number that gets reported

Equation (4) concerns policies. What a paper reports is a score $M(\theta)$: pass rate, exact match, win rate. Linearising about the endpoint, $\mathrm{Var}_{\mathrm{seed}}[M(\theta_T)] \approx \nabla M^\top S_T \nabla M$, which needs a matrix nobody can form for a real model. It is not needed.

Theorem 2 (Adjoint form of the reported spread)**.** *Define $b_T := \nabla M(\theta_T)$ and $b_s := A_s^\top b_{s+1}$ for $s = T-1, \dots, 0$. Then*

$$V_T^{lin} = \sum_{t=0}^{T-1} b_{t+1}^\top Q_t \, b_{t+1} = \frac{\eta^2}{P} \sum_{t=0}^{T-1} \mathrm{Var}_i\big(b_{t+1}^\top z_{t,i}\big), \tag{5}$$

the variance taken over prompts.

Proof sketch. $b_s = \Phi_{T,s}^\top \nabla M$ by induction; substitute the unrolled S_T. The second equality is $Q_t = \eta^2 \Sigma_t / P$ with $b^\top \Sigma_t b = \mathrm{Var}_i(b^\top z_{t,i})$, since Σ_t is the covariance of one prompt's contribution rather than of the batch mean. □

The equality is exact for V_T^{lin}, the variance of the metric linearised at $\bar{\theta}_T$. The quantity a practitioner wants is $\mathrm{Var}_{\mathrm{seed}}[M(\theta_T)]$, and the two agree to the order at which the linearisation holds; Section 7 is about when that agreement fails. Every term of (5) is one scalar per update, which a trainer can accumulate alongside the update it is already making.

What the backward pass costs depends on $A_s^\top$, and here the estimator family pays for itself.

Proposition 1 (The mean update is a gradient field)**.** *For any count-based estimator, $\bar{\boldsymbol{g}}(\theta) = \nabla_\theta \Psi(\theta)$ with $\Psi(\theta) = \mathbb{E}_x[\Lambda(p_x)]$ and $\Lambda' = \lambda$, where the per-prompt mean update is $\lambda(p_x, G)\nabla_\theta p_x$. Hence $\mathcal{J} = \nabla^2 \Psi$ is symmetric and $A_s^\top = A_s$.*

Because λ depends on θ only through p_x, $\lambda(p_x)\nabla p_x = \nabla \Lambda(p_x)$, and averaging over x commutes with the gradient. We call Ψ the *estimator potential*; it is not the objective J unless $\lambda \equiv 1$, and the gap is exactly the difficulty reweighting the estimator applies. For RLOO the potential is the pass rate itself. The practical consequence is that the backward pass needs no transpose and can be run with the same code path as the forward one.

3.2 Linearising the model instead of the run

Theorem 2 is exact in the model and first-order in the distance between runs, and it inherits its one run's blind spot: at a run that has saturated, $\nabla M(\theta_T) = 0$ and the forecast vanishes, wherever the other seeds went. The opposite linearisation is available. With $\boldsymbol{s}_0(x, y) := \nabla_\theta \log \pi_{\theta_0}(y \mid x)$ the score at the starting parameters, the *tangent policy*

$$\pi'_\vartheta(y \mid x) \;\propto\; \pi_{\theta_0}(y \mid x)\, \exp\big(\boldsymbol{s}_0(x, y)^\top \vartheta\big) \tag{6}$$

is an exponential family in the frozen scores. Updating ϑ with an estimator of (1) needs sampled responses, their rewards and inner products of frozen scores, and no pass through the model, so seeds can be *simulated* by the thousand with the sampling, the estimator's weights and the saturation all kept exactly.

Proposition 2 (When a simulated seed is a real one)**.** *If $\log \pi_\theta(y \mid x)$ is affine in θ up to a per-prompt normaliser, the run from θ_0 and the tangent run from $\vartheta_0 = 0$ have the same law, so $M(\theta_T)$ has the distribution of its simulated counterpart.*

Both are then Markov chains with one transition kernel. A softmax over one decision qualifies. A network does not: there a simulated run is exact in its first update and drifts as the scores move. This is the tangent-kernel approximation [Jacot et al., 2018, Malladi et al., 2023], used to draw many runs instead of predicting one.

Algorithm 1 The forecast, as a trainer would run it

train as usual, storing θ_t and η_t for each update
$b \leftarrow \nabla M(\theta_T)$ ▷ one rollout and backward pass on the eval set
$v \leftarrow 0$
for $t = T-1$ **down to** 0 **do**
  draw a batch at θ_t; let $\hat{g}_{t,i}$ be its per-prompt gradients
  $v \leftarrow v + (\eta_t^2/P)\,\widehat{\mathrm{Var}}_i\big(b^\top \hat{g}_{t,i}\big)$ ▷ one scalar
  if transport is needed **then**
    $\delta \leftarrow \epsilon\, b/\|b\|$
    $b \leftarrow b + \frac{\eta_t}{2}\cdot\frac{\|b\|}{\epsilon}\big(\bar{\boldsymbol{g}}(\theta_t+\delta) - \bar{\boldsymbol{g}}(\theta_t-\delta)\big)$ ▷ common random numbers
  end if
end for
return $\sqrt{v}$

4 Running it

What it costs, plainly. As written, the backward pass makes three passes per stored update: one batch for the injected term and two for the Hessian-vector product. That is about three times the compute of the training run, not a rounding error, and we would rather state it than let a reader discover it.

It is also usually unnecessary. Ablating the transport, holding b at $\nabla M(\theta_T)$ instead of carrying it back, changes the forecast by a median of 2% and the median error not at all over runs of ten and twenty-five updates. The reason is in the spectrum of A_t: most of it sits within 10^{-4} of one and ∇M points into that part, so $\Phi_{T,t}$ is close to the identity along the one direction the forecast needs. What we recommend is the cheap estimator,

$$\mathrm{Var}_{\mathrm{seed}}\big[M(\theta_T)\big] \;\approx\; \sum_t \frac{\eta_t^2}{P}\,\widehat{\mathrm{Var}}_i\big(\nabla M(\theta_T)^\top \hat{g}_{t,i}\big), \quad (7)$$

one extra pass per update and no Hessian-vector products, with the full recursion held in reserve for runs long enough, or curvature strong enough, that a per-update discrepancy of a percent compounds. Knowing which regime one is in is what the recursion is for.

A closed form with no run at all. Take (7) with $\Phi \approx I$ and a constant step, replace the projection $\mathrm{Var}_i(\nabla M^\top \hat{g}_{t,i})$ by the trace (2), and the run drops out entirely:

$$\sigma \;\propto\; \eta\sqrt{T/P}\,\sqrt{\tau_{\mathrm{b}} + \tau_{\mathrm{w}}/(G-1)}. \quad (8)$$

We call this the *frozen* forecast because it needs only the run length, the batch shape and the two noise traces, and is therefore available before any training happens.

It is written as a proportionality on purpose. The constant it is missing carries $\|\nabla M\|$, which is exactly the part that requires a trained model, and on our grid (8) overstates the measured deviation by a factor of about 84. That factor is not quite constant: it varies by 4.8× across the grid, leaving a relative error of 1.26× at the median once one common scale is removed. So (8) is useless for reporting an error bar and, by the argument of Section 6, entirely serviceable for sizing one, because a stopping threshold absorbs a constant and nothing absorbs the part that varies.

Both traces come from the microbatch gradients a trainer already accumulates: with blocks a holding disjoint prompts and sub-groups $b \in \{1,2\}$ independent given a prompt,

$$\begin{aligned} \operatorname{tr}\Sigma_{\mathrm{b}} &= \tfrac{P}{K}\big(\mathbb{E}[g_{a1}^\top g_{a2}] - \mathbb{E}[g_{a1}^\top g_{a'2}]_{a\neq a'}\big), \\ \operatorname{tr}\Sigma_{\mathrm{w}} &= \tfrac{P}{2K}\,\mathbb{E}\big[\|g_{a1}-g_{a2}\|^2\big]. \end{aligned} \quad (9)$$

The cost is $2K$ gradient buffers rather than one, no extra responses and no extra backward work. Estimating the total and subtracting the within-prompt part instead differences two large correlated quantities and is badly conditioned: on homogeneous pools it returns a negative τ_{b}. Against exactly known values over pools whose noise ratio spans a factor of 26, (9) recovers the optimal group size to a median error of 1.3%, while the subtractive form reports 6.1 where the truth is 9.4.

Section 6 finds that (8) does most of the work, which matters because it costs almost nothing.

5 Does the forecast work

We test where the answer can be checked exactly. The environments are small enough to enumerate every sequence, so $\bar{\boldsymbol{g}}$, the Fisher and the metric gradient are computed in closed form rather than sampled, and the forecast is compared against a spread measured from 47 independent seeds. The grid spans two policy shapes, two prompt-pool diversities, three difficulty bands, two group sizes and a range of step sizes: 192 settings in all, over which the measured spread itself varies by 46×.

The test is prospective. One run is trained, the forecast is computed from it alone, and only then are the remaining seeds trained. The anchor is excluded from the measured spread, since a predictor that is also one of its own targets is not independent of them.

Table 1 reports a median error of 1.10× and a worst case of 1.56×, with the forecast inside the measured 95% interval in 162 of 192 settings. The grid uses a pool of 24 prompts; the same 192 settings on a pool of 16, run as a replicate, give 1.10×, 1.50× and 154 of 192.

The recursion is what predicts divergence. Two cheaper stories are available for the full covariance, and both fail. Accumulation, the random-walk

grouped by	settings	median	worst	inside 95%
all settings	192	1.10×	1.56×	162/192
policy 3^3	96	1.10×	1.56×	82/96
policy 4^2	96	1.10×	1.38×	80/96
$T = 10$	96	1.10×	1.34×	86/96
$T = 25$	96	1.11×	1.56×	76/96
$P = 8$	96	1.12×	1.56×	78/96
$P = 32$	96	1.08×	1.53×	84/96
$\eta = 0.5$	96	1.10×	1.29×	92/96
$\eta = 1.5$	96	1.13×	1.56×	70/96

Table 1: Forecast against measured spread, prospectively. Error is the ratio in whichever direction exceeds one.

picture that ignores A_t entirely, is wrong by a factor of two on homogeneous prompt pools, which is where the objective's curvature is most consistent and the filtering therefore strongest; on heterogeneous pools it is wrong by three percent, which is what (4) predicts. A single effective timescale fails by two orders of magnitude. Measuring the spectrum of A_t says why neither works: it is not a contraction. The spectral radius exceeds one throughout, by up to 2% early and decaying as the policy improves, and only 31–51% of directions contract. Divergence stays bounded because expansion is slow relative to run length, not because every direction shrinks.

The filtering is caused by learning. Replacing rewards with coin flips, which leaves the rollouts, the batch composition and the step sizes untouched, turns a divergence trace that ends at 0.14 of where it started into one that grows by a factor of 90.5 over eighty updates, a gap of 671× at the end. The learning condition also moves further per update, so what brings those runs back together is not smaller steps.

When the run learns. The settings above gain at most 0.036 in pass rate; the pretrained runs of Section 7 gain up to 0.28. Raising the step size until enumerable runs gain as much, on 384 further settings with exact inputs, each graded against 500 seeds and with its predictions recorded first, breaks the forecast. On the 354 settings gaining at least 0.05 it is more than 3× off in 44.1%, nearly always short, and by up to 1715× once the policy has collapsed onto one sequence. Seeds simulated in the tangent model at θ_0, with no run trained, are within 1.5× in 74.9% of those settings and beyond 3× in 4.8%; the deviation over ten real seeds manages 73.7% and 11.3%, and the forecast 44.4%. Four of the six parts we had recorded failed, three on one clause. A tabular policy and its tangent model span the same family, so this tier flatters the surrogate, and on a network it is untested.

6 What the forecast is for

A predicted standard deviation is not a result on its own. Nobody trains a model in order to learn how much its score would move under a different seed; they want to know whether one recipe beats another, and the deviation is what says how many runs settle it.

6.1 The decision

Two configurations are to be ordered. A practitioner trains each for a seed, looks at the lead, and either calls it or trains another seed of each. Stopping when the lead $\bar{D}_k$ after k seeds per arm clears z standard errors,

$$|\bar{D}_k| \;\geq\; z\,\frac{\sigma_D}{\sqrt{k}}, \qquad \sigma_D = \sqrt{\sigma_A^2 + \sigma_B^2}, \tag{10}$$

is the rule this instantiates. It needs σ_D, and that is the difficulty: at $k = 1$ there is no empirical deviation to put there. The remaining choices are to fix k in advance, which is what is normally done, or to spend seeds building a variance estimate before the comparison can begin. Sections 3 and 4 supply σ_A and σ_B from the first run of each arm, so (10) is available from $k = 1$.

Multiplying every σ by a common factor is undone by dividing z by it, so if z can be tuned, only the *relative* deviation across configurations matters. Tuning z needs comparisons whose outcome is known; Section 6.4 asks what remains without them.

6.2 Setup

The scores are the 48 per-seed scores already stored for each of the 192 configurations of Section 5; no new training is done, and every policy sees the same seeds in the same order, so the comparison is paired. We compare configurations sharing a task.

The true ordering has to come from somewhere, and reading it from seeds a policy is also shown would correlate the policy's evidence with the answer it is graded against, flattering whichever policy looks at the most seeds. So 32 of the 48 seeds are reserved: the ordering is fixed on those and no policy may draw one. The 1020 pairs whose reserved-seed ordering is unambiguous at four standard errors are kept, so a disagreement is a mistake rather than a coin landing the other way up, and policies draw from the remaining 16. Each pair is replayed under 400 random orderings.

A policy is a curve rather than a point, since z trades runs against mistakes, and reading every curve at a common mistake rate is what makes two comparable. We report the runs each needs to hold its mistake rate at 0.1%, both arms counted.

policy	runs	vs. adjoint	95%
fixed budget, no stopping rule	30.41	9.57	[7.24, 10.56]
deviation from the seeds in hand	6.52	2.13	[1.98, 2.38]
one deviation for every recipe	6.04	1.93	[1.45, 2.28]
frozen forecast, before any run	3.34	1.09	[1.02, 1.14]
adjoint forecast, from one run	3.07	1.00	[1.00, 1.00]
measured deviation	3.16	1.02	[0.99, 1.05]

Table 2: Runs needed to hold the mistake rate at 0.1% over 1020 comparisons, both arms counted, with 95% intervals from resampling the comparisons. The forecast is not better than knowing the deviation; it is not measurably worse, which is the point.

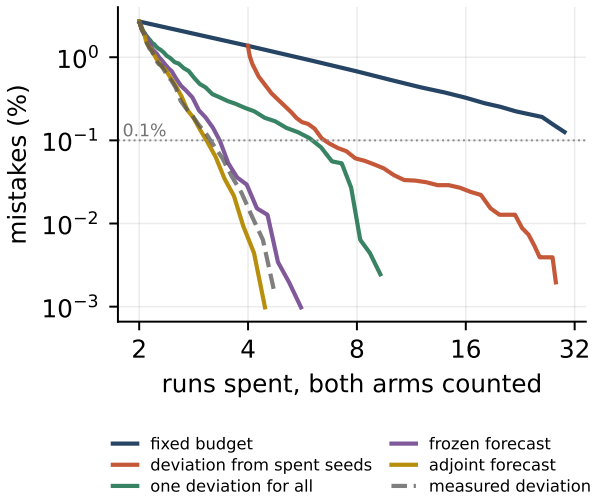


Figure 1: Mistakes against runs spent. A policy is a curve, not a point, since the stopping threshold trades one against the other, and what separates two policies is how far left a curve sits at a fixed height. At 0.1% the forecasts cross near 3.07 runs and a fixed budget near 30.4. The rule driven by spent seeds flattens on the left because it cannot stop before two seeds per arm however large the lead.

6.3 Result, with z read in sample

The adjoint forecast reaches 3.07 runs and a rule handed the realised deviation 3.16, the same to within resampling error (1.02× [0.99, 1.05]). A fixed budget needs 30.4 runs, 9.57× [7.24, 10.56] more, and estimating the deviation from spent seeds 2.13× more, mostly because that rule cannot stop at one seed per arm. An earlier version graded against all 48 seeds, a policy's own included; that leak favours the baselines that draw the most seeds and cut the fixed-budget gap to 4.76×. The closed form (8), with no stored run and no backward pass, reaches 3.34 runs, within 1.09× of the adjoint: most of the saving does not need Theorem 2.

6.4 Setting the threshold before the comparison

Those numbers read every policy at the threshold meeting the target on the comparisons it is graded on: even-handed, but not something a practitioner can do. Table 3 repeats the benchmark two ways one could. Either the threshold, or for a fixed budget the seed count, is the cheapest meeting the target on the other eleven tasks and is applied unchanged to the twelfth; or there is no tuning, and the rule stops once the lead clears the one-sided normal quantile of the target. Intervals resample the twelve tasks, since comparisons inside a task share arms.

	target 1%		target 0.1%	
threshold set on other tasks	runs	mistakes	runs	mistakes
fixed budget	6.00	0.91%	32.00	0.00%
one deviation for every recipe	2.47	0.98%	6.24	0.12%
deviation from spent seeds	4.05	1.02%	6.66	0.10%
closed form, no run	2.30	0.96%	3.48	0.08%
closed form, one scale from other tasks	2.29	0.98%	3.47	0.08%
adjoint forecast	2.23	0.98%	3.14	0.10%
measured deviation	2.22	0.95%	3.22	0.09%
textbook threshold, no tuning				
one deviation for every recipe	4.36	0.22%	5.77	0.11%
closed form, no run	32.00	0.00%	32.00	0.00%
closed form, one scale from other tasks	4.64	0.01%	6.23	0.00%
adjoint forecast	4.37	0.00%	5.96	0.00%
fixed budget, 24-seed cap, 3 splits	4.0–4.0		15.3–23.3	

Table 3: Runs per comparison and realised mistake rate over 1020 comparisons with no threshold read off the comparisons graded. Upper block: threshold fitted on the other eleven tasks. Lower block: the textbook threshold. At 0.1% the fixed budget sits at the 16-seed cap; the last row extrapolates it without one.

Thresholds chosen on other tasks transfer: realised mistake rates land near the target. At 0.1% the adjoint forecast needs 3.14 runs; the fixed budget sits at its cap here, so we also split the seeds 24/24, where it can be measured below a cap: over 3 splits it needs 15.3 to 23.3 runs, 5.6 to 8.2× the forecast. Against the closed form it is 1.11× [0.97, 1.28] with exact noise traces and 1.18× with traces estimated from one run: once z can be learned on other tasks, the stored run adds little to a forecast that only has to rank.

At the textbook threshold the uncalibrated closed form never stops, its scale off by 84×; given one scale learned on other tasks it decides in 4.64 runs with 0.013% mistakes at a 1% target, and the adjoint forecast in 4.37 with 0.001%. The stored run's scale stands in for that one learned factor, and both textbook rules spend about twice the tuned runs. Keeping near-ties (two standard errors instead of four) puts every policy at the cap at 0.1%; no policy settles near-ties with the seeds we have.

The other use of a deviation is choosing: run several configurations once and keep the one with the most gain per unit of noise. With seeds split three ways, never reused across roles, the choice keeps 66% of the best achievable value with one deviation for every recipe and 75% with the forecast, against 77% with the measured deviation: the forecast recovers 8.8 points [1.7, 15.6] of 10.2. Choosing has little headroom because one pilot's gain is itself noisy.

6.5 How wrong the forecast may be

On a pretrained model the forecast is accurate only in some regimes (Section 7), so we corrupt it on purpose and rerun the benchmark: a lognormal multiplicative error of controlled median magnitude, and separately a systematic scale error.

The scale error does nothing here. Over 100-fold changes in scale the runs needed are identical to the digits we print, a factor of 1.000, because z absorbs it exactly as (10) says. That holds only because this benchmark reads z off the graded comparisons; with the textbook threshold a scale error is not absorbed, which is why the closed form never stops in Table 3. Calibration is free only to someone who already has comparisons with known outcomes to set z on.

Relative error is what costs, and it costs less than we expected. At the 0.1% target the runs needed rise from 3.07 at the measured accuracy to 5.4 at a median error of 1.5×, 9.5 at 2×, 13.3 at 2.5× and 18.7 at 4×, and never reach the fixed budget's 30.4 anywhere in the range we corrupted, which extends to 12×. At a looser 1% target, where a fixed budget needs only 5.61 runs, the break-even is 4.2×.

That break-even, measured with z read in sample, is the statement of scope. The 4 pre-specified regimes of Section 7, at $1.66, 2.13, 2.80, 1.24$, sit inside it. Two of our pretrained results do not: the task on which most runs learn to answer one constant, missed by 25×, and Qwen2.5-7B under Adam on the same task, short by between 3.1 and 17, a range reaching well past it. Where the forecast misses that badly, a forecast-driven stopping rule would stop too early.

7 On pretrained models

Run end to end on Qwen2.5-0.5B with LoRA, the cheap estimator (7) is a factor of 25 short of the measured spread and the transported one 16. On Qwen2.5-7B under Adam, with the recursion lifted to the optimiser state $z = (\theta, m, v)$, it is short by between 3.1 and 17. On the exact models the lift gives 1.10× with Adam's moments warmed, but started cold and without bias correction, as these runs were, its median error there is 2.40× and its worst 549×. Section 5 reproduces the shortfall on exact models once they learn, so it does not need a pretrained network.

Accuracy is regime-dependent, not uniformly poor. On the corpus above the reward groups go unanimous within five updates, yet its measured seed spread is the largest we report. Most runs there learn to answer one constant: the stored run says "1" to every prompt, and which constant a run falls into explains 94% of the spread. So we built four corpora at pre-specified mean live-group probabilities and ran the prospective test on each. The errors are $1.66, 2.13, 2.80, 1.24$; the best, on the band at 0.85, meets a criterion of 1.5× declared before the sweep ran, forecasting 0.0205 against a measured 0.0253 $[0.0105, 0.0337]$ on a gain of 0.275. So local covariance forecasting can be quantitatively accurate on a pretrained model, though not reliably so. Nor is collapse the cause: at 0.60, where the forecast is 2.80× too large, runs barely collapse, refuting two predictions we recorded beforehand.

All four sit inside the 4.2× break-even of Section 6, measured with z read in sample; the unanimous corpus and 7B under Adam do not.

The sizing benchmark at this scale. Across 6 recipes on one corpus (102 runs) the forecast holds up, at a median error of 1.61× and a worst of 2.75×. The benchmark does not: only 4 of 15 pairs are ordered unambiguously at 2 standard errors, most being genuine near-ties at sixteen seeds. On four comparisons a stopping rule beats a fixed budget by 1.41 $[1.28, 1.65]$ and nothing else separates. The 9.57× of Section 6 is therefore not demonstrated at this scale and we do not claim it.

Two things the sweep falsified. It does not support a monotone relationship between stochastic support and accuracy: the ordering of errors across the four bands does not follow the ordering of their live-group probabilities. And a one-run curvature diagnostic, calibrated on the exact models to refuse forecasts whose runs are not small perturbations of each other, does not transfer: it refuses every pretrained run, including the accurate one.

8 Related work

Run-to-run variance. That RL results move under reseeding is old news in deep RL [Henderson et al., 2017] and documented again for language-model reasoning [Hochlehnert et al., 2025]. Those papers establish the size of the problem by replication, and Colas et al. [2018] turn it into a power analysis that sizes the seed count from a pilot sample of seeds, which is the practice our sizing experiment replaces. Agarwal et al. [2021] prescribe how the resulting samples should be summarised. All of it presumes the runs exist. Heineman et al. [2025] estimate a benchmark's noise within one run, from checkpoint to checkpoint; ours is the across-seed spread of the training procedure. Practice for benchmark error bars treats the evaluation set as the only source of randomness [Miller, 2024], which is the right interval for a fixed model and the wrong one for a training procedure; Bouthillier et al. [2021] argue

the pipeline's own randomness belongs in the interval and estimate it by replication. We estimate the same quantity without replicating, and our real-model measurements subtract the binomial term explicitly.

Deciding when to stop repeating. Stopping a comparison once one arm is convincingly ahead is sequential analysis [Wald, 1945], and its machine-learning form is racing [Maron and Moore, 1993, Even-Dar et al., 2006], sharpened by using an empirical variance in the bound [Audibert et al., 2009] and scaled to hyperparameter search by allocating a budget across candidates [Li et al., 2017]. Section 6 uses the simplest rule of that family deliberately: the contribution is not a better stopping rule but the quantity the rule consumes. Every method there takes its variance from samples already drawn, which is sound when a sample is a validation example and unaffordable when a sample is a training run. At one run per arm those bounds are vacuous, and the experiment shows that the resulting floor, not the shape of the bound, is what a comparison pays.

Stochastic dynamics of training. Treating stochastic optimisation as a process whose covariance is set by gradient noise against curvature is standard [Mandt et al., 2017], and (4) is not new as mathematics. What we add is the RLVR-specific measurement of Q_t through the two-level decomposition, the finite-time rather than stationary treatment, and the observation that the filter, not the size of the noise, is what governs reproducibility here. Wang et al. [2026] reach a filtering picture from the opposite side, reporting near-linear evolution of weights and log-probabilities; their object is one run's trajectory, ours the covariance between runs differing only in seed. Closest to us, Ghosh et al. [2026] reduce GRPO to a scalar damped oscillator with a stationary fluctuation scaling as $1/G$; we keep the full covariance, at finite time, and test it on seeds trained after the forecast was frozen.

Adjoint sensitivity. Theorem 2 applies the adjoint method [Pontryagin et al., 1962, Chen et al., 2018] to a covariance rather than a loss.

Entropy collapse. That RLVR drives policy entropy down sharply is documented [Cui et al., 2025]. By (5) it has a second consequence: a unanimous group has zero count-based advantage, so a collapsed policy injects $\Sigma_t = 0$ and stops accumulating seed uncertainty.

Batch size and gradient noise. The gradient noise scale predicts the batch size beyond which data parallelism stops buying speed [McCandlish et al., 2018, Shallue et al., 2018]. Both concern a single level of sampling. RLVR samples prompts and then responses within each prompt, and the resulting two-level scale has no counterpart there; the decomposition itself is the classical two-stage sampling result [Cochran, 1977]. Zhou et al. [2026] concurrently derive an optimal group size from the mean-squared error of the GRPO gradient as a U-statistic; ours minimises injected noise at fixed rollout budget, and the two remain uncompared.

Allocating rollouts. Recent methods vary responses per prompt within a batch [Nguyen et al., 2026] or adapt the batch size during training [Park, 2026]. Those allocate samples *within* one run; the budget here is across runs of the same recipe.

9 Limitations

The sizing benchmark. The stopping rule (10) is not anytime-valid. Testing after every seed and stopping on the first crossing inflates the error of a fixed-sample test, so its nominal z carries no coverage guarantee and we claim none. What we report is the mistake rate the rule realises, measured against an ordering fixed on seeds no policy may draw. Every policy is graded the same way and separated at a common realised rate, so the comparison does not rest on the boundary being calibrated; a sequentially valid boundary would move all of them together, which we expect but have not verified. Pairs whose ordering is ambiguous are excluded, so the benchmark says nothing about near-ties, where the honest answer is to keep training whatever rule is in use.

Scale. The largest model is a 4-bit 7B with 10 seeds, on one laptop, and the exact tier where the forecast is validated is the furthest from a frontier run.

What the forecast costs. Estimating the injected term from the eight prompts a run trains on gives two forecasts 681× apart, because only about a tenth of groups are not unanimous. Raising that to 32 prompts brings disjoint halves to within 5.5× and costs 141 minutes against the 260 that training all 10 seeds costs. A diagnostic consuming half the experiment it replaces is not yet a diagnostic at this scale.

Claims we withdrew. A predicted square-root relation between policy divergence and outcome spread is not supported by our data. Nor is a short memory horizon: in 254 of 288 settings more of the final variance was injected in the first half of the run. Token-level reshaping, length normalisation and clipping are outside (1).

How this work was done. The author chose the research questions, decided which experiments to run, judged the results, decided which claims to keep, correct or withdraw, and derived and checked parts of the theory. AI coding agents (Claude Code) wrote most of the code and much of the text, ran the experiment sweeps under that direction, and in a later revision found several of the errors this version corrects. Every number in the tables and in the generated macros is computed from the stored run records released with the code.

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